

STUDENT RESPONSE SHEET - RAHUL VERMA

ANSWER BOOKLET - DESCRIPTIVE GRADING

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Answer 1:

Stack is a list where you push elements at the top. Queue is where you enqueue elements at the rear and dequeue them at the front. Stack is FIFO (First In First Out) and Queue is LIFO (Last In First Out). No wait, Stack is LIFO (Last In First Out) and Queue is FIFO. Real-world example: Stack of plates, Queue at supermarket. Stack is used to store function parameters.

Answer 2:

TCP is connection oriented. UDP is connectionless. TCP guarantees packets are delivered, but UDP packages can be lost. TCP is slow because of handshake overhead. UDP is fast for games. TCP works on 3 way handshake. UDP has no handshake.

Answer 3:

Binary search is a method to find an element in a sorted list. It goes to the middle item, if it matches it returns. If not, it searches recursively. Python code:

```
def search(arr, x):  
    for i in range(len(arr)):  
        if arr[i] == x: return i  
    return -1
```

Complexity of this search code is $O(n)$ because it checks everything in a loop. So this is linear search. But binary search is $O(\log n)$.

Answer 4:

Bayes theorem formula: $P(A|B) = P(B|A)P(A)/P(B)$. The test accuracy is 99%. Since the patient tests positive and the test accuracy is 99%, the probability that they have the disease is 99% because the test is very accurate.